

Conditional Parametric Model for Sensitivity Factors in LV Grids: A Privacy-Preserving Approach

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Abstract—The deployment of smart metering technologies in the low voltage (LV) grid created conditions for the application of data-driven monitoring and control functions. However, data privacy regulation and consumers’ aversion to data sharing may compromise data exchange between utility and customers. This work presents a data-driven method, based on smart meter data, to estimate linear sensitivity factors for three-phase unbalanced LV grids, which combines a privacy-preserving protocol and varying coefficients linear regression. The proposed method enables centralized and peer-to-peer learning of the sensitivity factors. One of the applications for the sensitivity factors is demonstrated by solving voltage violations or computing operating envelopes in a LV grid without resorting to its network topology or electrical parameters.

Index Terms—Low voltage, sensitivity factors, privacy-preserving, voltage control, smart meters, conditional model.

I. INTRODUCTION

The transformation of the electric power system to safely host more renewable energy sources (RES), new loads (e.g., electric vehicles, EV) and consumer behavior, requires a new distribution network operation paradigm, taking advantage of flexibility (storage, demand response, RES curtailment, etc.) and digital solutions that improve real-time observability and control. LV networks (LV) are key in this new operation paradigm, which face a steadily growing penetration of small-scale PV, storage and EV, remain a major bottleneck due to the following reasons: (a) lack of sensors and monitoring equipment, as well as a reliable, close to real-time and robust data communication between the smart meters and distribution system operator (DSO) head-end, motivating the development of data-driven state estimation functions with frugal data requirements [1]; (b) reduced automation capability, with fuse-based protection or manually operated breakers, whereas storage and smart appliances flexibility could provide strategic

grid support services [2]; (c) continued RES integration and storage at consumer premises, on top of an unknown or outdated topological and electrical characterization, impelling the development of data-driven algorithms to estimate topology and electrical parameters [3], and behind-the-meter installed capacity [4].

This lack of a reliable and/or updated characterization of the network’s assets is a big constraint to apply conventional analytical techniques to the LV grid control and management, such as optimal power flow (OPF) and analytical sensitivity factors [5] (i.e., linearized dependencies between voltage magnitudes/current and nodal active/reactive power injections), since accurate information about network topology and electrical parameters is essential. An alternative is to apply data-driven methods to estimate sensitivity factors exclusively from active power and voltage measured by smart meters, without the need to have the grid model. This idea is not new and it was already proposed by different authors, namely: linearized AC power flow (PF) through the combination of partial least squares and Bayesian linear regression [6]; a noise robust data-driven linearized AC PF formulated as an error-in-variables problem solved with an iterative linearly constrained quadratic programming algorithm [7]; recursive weighted partial least squares using only synchronized voltage and active power injection measurements collected from phasor measurement units [8]. These data-driven sensitivity factors can be integrated into the model-free predictive control approaches for Volt-Var optimization [9].

For LV grids in particular, Weckx et al. proposed a linear regression method to estimate sensitivity factors between active power and voltage magnitude from smart meter data [10]. However, their method has two main limitations: (a) assume that access to private electrical energy consumption data is not a barrier; (b) grid operating regimes are identified manually and with empirical rules. Mugnier et al. proposed a method based on a system of linear equations solved with generalized least squares to calculate voltage sensitivity factors solely based on nodal power injections and voltage magnitude [11]. Silva et al. estimated voltage sensitivity factors using a ridge regression estimator, which are adaptive in time via a direc-

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tional forgetting scheme (i.e., older data is only discarded only in the directions where new information is available) [12]. Both works do not address data privacy and adaptability of sensitivity factors to the varying operating point. Moreover, it is important to note that adaptability to operating point changes cannot be achieved by just placing more weight on recent measurements (as proposed in [12]) since this only means coefficients changing in time (which is adequate for handling distributional shift) and not conditioned by external variables (or operating regimes) – see [13] for more details.

In terms of smart meter data privacy and sharing, the literature is divided in two groups: (a) monetary incentive mechanisms based on game theory that trade-off benefits from using data, privacy loss and consumers' willingness-to-participate [14]; (b) federated learning applied to load forecasting [15], and privacy-preserving techniques to enable different levels of spatial and temporal aggregation of individual smart meter data [16], [17]. The interested reader is referred to [18] for a critical review of privacy-preserving techniques for time series data.

Compared to existing literature, the present paper brings the following original contributions: (i) collaborative encryption protocol that ensures data protection during calculations of sensitivity factors with smart meter data (i.e., personal data) and the lossless recovery of the original data, being flexible enough to accommodate architectures with centralized and peer-to-peer data exchange – existing literature is mainly focused on smart meter data/information aggregation [16], [17]; (ii) explores varying coefficient linear regression theory (see [19]) to automatically handle different operating conditions of the LV grid. Moreover, the proposed mathematical framework can be combined with other approaches from the state-of-the-art, such as the directional forgetting scheme from [12] or the generalized least squares from [11], delivering a more robust data-driven estimator.

The remainder of this paper is organized as follows: Section II presents the formulation of the varying coefficients linear regression model and Section III the privacy-preserving protocol for handling smart meter data; Section IV presents the case-studies and discusses numerical results; Section V presents the final conclusions and topics for future work.

II. CONDITIONAL PARAMETRIC MODEL FOR SENSITIVITY FACTORS

This section formulates a conditional parametric model (i.e., varying coefficients) to extract, exclusively based on smart meter data, linear relations (sensitivity factors) between voltage magnitudes and active power injections.

A. Data-driven voltage sensitivity factors

From the voltage and active power data measured by smart meters spread across a LV grid, it is possible to compute sensitivity factors that estimate a new voltage magnitude after an active power control set-point in any grid node, i.e.,

$$V_i^{new} = V_i^{old} + \sum_{j=1}^n S_{i,j} \cdot \Delta P_j + \sum_{p=1}^3 S_{i,p}^{SS} \cdot \Delta V_p^{SS}, \quad (1)$$

where V_i^{new} and V_i^{old} are the new and the previous voltage for the connection point i , respectively, $S_{i,j}$ is the sensitivity of point i voltage in respect to point j power variation, ΔP_j is the active power variation in the connection point j , $S_{i,p}^{SS}$ is the voltage sensitivity of point i in respect to the secondary substation (SS) voltage variation in phase p , ΔV_p^{SS} . With this model, not only it is unnecessary to know the connection phase but also this is an information that can be extracted from the model fitting (i.e., $S_{i,p}^{SS}$ will be very close to 1 if point i is connected to phase p , and close to zero otherwise).

Note that the model does not consider reactive power measurements as covariates. The main reasons are: (i) reactive power consumption by LV consumers is considered to be low; (ii) LV feeders are known to have a large R/X ratio, diminishing the effect of reactive power in voltage. However, if available and relevant, reactive power measurements can be easily integrated into the model.

Regarding the voltage oscillations at the SS, the proposed approach assumes that the influence from active power control of LV grid resources is marginal, turning ΔV_p^{SS} into an input of the model. Moreover, using ΔV_p^{SS} as a covariate also enables the integration of control actions at the SS level, such as on-load tap-changers transformers.

B. Constant sensitivity factors

As formulated in [11], a sensitivity factor expresses the voltage magnitude variation in node i when the active power injection in node j is affected (i.e., $S_{i,j} = \Delta V_i / \Delta P_j$), and can be extracted from voltage and active power measurements via linear regression.

Let $\Delta V_{i,t}$ be the voltage variation, and $\Delta P_{i,t}$ the active power variation, from point i at time t , $i \in \{1, \dots, n\}$, $t \in \{1, \dots, T\}$. Then, $\{\Delta V_{i,t}\}_{t=1}^T$ follows a linear regression model with covariates $\{\mathbf{x}_t\}_{t=1}^T = \{[1, \Delta P_{1,t}, \Delta P_{2,t}, \dots, \Delta P_{n,t}]^\top\}_{t=1}^T$ when

$$\Delta V_{i,t} = \mathbf{x}_t^\top \cdot \mathbf{B}^{(i)} + \epsilon_{i,t}, \quad \forall t, \quad (2)$$

where $\mathbf{B}^{(i)} \in \mathbb{R}^n$ represents the coefficient vector, whose elements are the intercept and the sensitivity factors S ; and $\epsilon_{i,t}$ denotes a white noise that is independent and identically distributed with mean zero.

The coefficient vector can be determined by the typical ordinary least squares or by variants such as the generalized least squares and LASSO. Despite the obvious benefits of using a purely linear model to determine the sensitivity factors, there may be a concern about the extent of such assumption. In that regard, since these sensitivity factors can vary with the grid operating conditions (as showed in [10]), but are linear around each operating point, a conditional parametric model is proposed in the next subsection so that the linear factors (i.e., coefficient matrix $\mathbf{B}$) become a smooth function of some exogenous variables (e.g., MV/LV transformer load).

C. Linear regression with varying coefficients

In a varying-coefficient linear regression, the coefficients are allowed to change smoothly with the value of exogenous

variables, called “effect modifiers” [19]. For this problem, candidate exogenous variables are MV/LV transformer load, voltage in secondary substation, or irradiance (related to PV generation) in the network location.

Considering the random variable y and a group of exogenous variables $\mathbf{u}$, the varying-coefficients model assumes the following form

$$y = \mathbf{x}^\top \cdot \mathbf{B}(\mathbf{u}) + \varepsilon, \quad (3)$$

where $\mathbf{B}(\cdot)$ is a vector of coefficient functions to be estimated. Regardless of the number of covariates, the dimension of $\mathbf{u}$ must remain low for practical purposes. Then, for a number of distinct points of the exogenous variables (e.g., between 0 and the rated power of the transformer with 10 kW increments), functions $\mathbf{B}(\cdot)$ are estimated by fitting locally a polynomial around each one of these *fitting points*.

To be more specific, and following the notation of [20], let $\mathbf{B}_j(\cdot)$ be the j^{th} element of $\mathbf{B}(\cdot)$ and $\mathbf{p}_d(\mathbf{u})$ be a column vector of terms in the corresponding d -order polynomial evaluated at $\mathbf{u}$. For instance, if $\mathbf{u} = [u_1, u_2]^\top$, then $\mathbf{p}_2(\mathbf{u}) = [1, u_1, u_2, u_1^2, u_1 \cdot u_2, u_2^2]^\top$. Moreover, let consider $\mathbf{x} = [x_1, \dots, x_p]^\top$, where p is the number of covariates, and $\mathbf{z}^\top = [x_1 \cdot \mathbf{p}_{d(1)}^\top(\mathbf{u}), \dots, x_j \cdot \mathbf{p}_{d(j)}^\top(\mathbf{u}), \dots, x_p \cdot \mathbf{p}_{d(p)}^\top(\mathbf{u})]$.

For the sake of illustration, for two covariates $\mathbf{x} = [x_1, x_2]^\top$ and one exogenous variable $\mathbf{u} = u_1$, we have:

- (a) 0-order polynomial: $\mathbf{p}_0(\mathbf{u}) = [1]$, $\mathbf{z}^\top = [x_1, x_2]$.
- (b) 1-order polynomial: $\mathbf{p}_1(\mathbf{u}) = [1, u_1]^\top$, $\mathbf{z}^\top = [x_1, x_1 \cdot u_1, x_2, x_2 \cdot u_1]$.
- (c) 2-order polynomial: $\mathbf{p}_2(\mathbf{u}) = [1, u_1, u_1^2]^\top$, $\mathbf{z}^\top = [x_1, x_1 \cdot u_1, x_1 \cdot u_1^2, x_2, x_2 \cdot u_1, x_2 \cdot u_1^2]$.

Now, considering $\hat{\phi}^\top(\mathbf{u}) = [\hat{\phi}_1^\top(\mathbf{u}), \dots, \hat{\phi}_j^\top(\mathbf{u}), \dots, \hat{\phi}_p^\top(\mathbf{u})]$ where $\hat{\phi}_j(\mathbf{u})$ is a column vector of local estimates at $\mathbf{u}$ (i.e., *fitting points*) corresponding to $x_j \cdot \mathbf{p}_{d(j)}(\mathbf{u})$, the linear model becomes

$$\mathbf{y} = \mathbf{z}^\top \cdot \hat{\phi}(\mathbf{u}) + \epsilon. \quad (4)$$

If $\mathbf{u}_0$ denotes a particular *fitting point*, a local estimation can be achieved using the classical weighted least squares (WLS):

$$\hat{\phi}(\mathbf{u}_0) = \underset{\phi_u}{\operatorname{argmin}} \sum_{t=1}^T \omega(\mathbf{u}_t) (y_t - \mathbf{z}_t^\top \cdot \phi_u)^2, \quad (5)$$

where the weights $\omega(\mathbf{u})$ are computed as

$$\omega(\mathbf{u}) = K \left(\frac{\|\mathbf{u}_0 - \mathbf{u}\|}{h(\mathbf{u})} \right), \quad (6)$$

with $\|\cdot\|$ denoting the Euclidean norm and $h(\mathbf{u})$ the bandwidth of the tricube kernel $K(\cdot)$ from (7),

$$K(d) = \begin{cases} (1 - d^3)^3 & \text{if } |d| \leq 1, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

Since the exogenous variable may be irregularly spaced, a fixed bandwidth could lead to some local fittings based on many points while others could use very few points. Thus, instead of using a fixed bandwidth value, it is preferable to

use a bandwidth that varies according to the density of data, the so-called nearest neighbor bandwidth:

- 1) Calculate the Euclidean norm for all T sample points.
- 2) Find the smallest integer k corresponding to $\alpha \cdot T$ (α is a user's defined hyper-parameter).
- 3) Sort the Euclidean distance by ascending order and set the bandwidth equal to the k^{th} element.

Going back to the local coefficients obtained in (5), the j^{th} element of the set of functions $\mathbf{B}(\mathbf{u})$ is then estimated by

$$\hat{B}_j(\mathbf{u}) = \mathbf{p}_{d(j)}^\top(\mathbf{u}) \cdot \hat{\phi}_j(\mathbf{u}). \quad (8)$$

For a case with two covariates and one exogenous variable,

- (a) 0-order polynomial: $\hat{B}_1(\mathbf{u}) = \hat{\phi}_{0,1}$, $\hat{B}_2(\mathbf{u}) = \hat{\phi}_{0,2}$, with $\hat{\phi}(\mathbf{u}) = [\hat{\phi}_{0,1}, \hat{\phi}_{0,2}]^\top$.
- (b) 1-order polynomial: $\hat{B}_1(\mathbf{u}) = \hat{\phi}_{0,1} + \hat{\phi}_{1,1} \cdot u_1$, $\hat{B}_2(\mathbf{u}) = \hat{\phi}_{0,2} + \hat{\phi}_{1,2} \cdot u_1$, with $\hat{\phi}(\mathbf{u}) = [\hat{\phi}_{0,1}, \hat{\phi}_{1,1}, \hat{\phi}_{0,2}, \hat{\phi}_{1,2}]^\top$.
- (c) 2-order polynomial: $\hat{B}_1(\mathbf{u}) = \hat{\phi}_{0,1} + \hat{\phi}_{1,1} \cdot u_1 + \hat{\phi}_{2,1} \cdot u_1^2$, $\hat{B}_2(\mathbf{u}) = \hat{\phi}_{0,2} + \hat{\phi}_{1,2} \cdot u_1 + \hat{\phi}_{2,2} \cdot u_1^2$.

Finally, after building the coefficient functions of the conditional model for the multiple *fitting points* (range of the exogenous variable), the sensitivity factors functions, for a given operating point, can be obtained by linear interpolation.

III. PRIVACY-PRESERVING PROTOCOL

In order to obtain the sensitivity factors, all the consumers have to contribute with data, namely historical voltage magnitude and injected/consumed active power. To overcome any privacy issues that could arise, inspired by [21], a collaborative protocol for linear models is proposed and that accepts two communication schemes, central node and peer-to-peer (P2P).

A. Distributed linear regression

As illustrated in Fig. 1, when considering a linear regression to model the relationship between voltage magnitude variation and active power variation, with varying coefficients as in (4), each time series is collected by a specific agent, i.e., $\mathbf{Y} = [\mathbf{Y}_1, \dots, \mathbf{Y}_n]$ and $\mathbf{Z} = [\mathbf{Z}_1, \dots, \mathbf{Z}_n]$, where $\mathbf{Y}_i$ and $\mathbf{Z}_i$ denote the target and covariate matrix for the i^{th} agent, respectively. Furthermore, $\Phi(\mathbf{u})$ is split by rows. This allows to express the weighted multivariate least-squares solution,

$$\underset{\Phi}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{W}^{\frac{1}{2}}(\mathbf{Y} - \mathbf{Z}\Phi)\|^2 = (\mathbf{Z}^\top \mathbf{W} \mathbf{Z})^{-1} (\mathbf{Z}^\top \mathbf{W} \mathbf{Y}), \quad (9)$$

where $\mathbf{W}$ is the diagonal matrix of weights, as

$$\underbrace{([\mathbf{Z}_1, \dots, \mathbf{Z}_n]^\top \mathbf{W} [\mathbf{Z}_1, \dots, \mathbf{Z}_n])^{-1}}_{\begin{bmatrix} \mathbf{Z}_1^\top \mathbf{W} \mathbf{Z}_1 & \dots & \mathbf{Z}_1^\top \mathbf{W} \mathbf{Z}_n \\ \vdots & \ddots & \vdots \\ \mathbf{Z}_n^\top \mathbf{W} \mathbf{Z}_1 & \dots & \mathbf{Z}_n^\top \mathbf{W} \mathbf{Z}_n \end{bmatrix}^{-1}} \underbrace{([\mathbf{Z}_1, \dots, \mathbf{Z}_n]^\top \mathbf{W} [\mathbf{Y}_1, \dots, \mathbf{Y}_n])}_{\begin{bmatrix} \mathbf{Z}_1^\top \mathbf{W} \mathbf{Y}_1 & \dots & \mathbf{Z}_1^\top \mathbf{W} \mathbf{Y}_n \\ \vdots & \ddots & \vdots \\ \mathbf{Z}_n^\top \mathbf{W} \mathbf{Y}_1 & \dots & \mathbf{Z}_n^\top \mathbf{W} \mathbf{Y}_n \end{bmatrix}}$$

that reveals the coefficients' estimation requires the computation of $\mathbf{Z}_i^\top \mathbf{W} \mathbf{Z}_j$ and $\mathbf{Z}_i^\top \mathbf{W} \mathbf{Y}_j$, $i \neq j$.

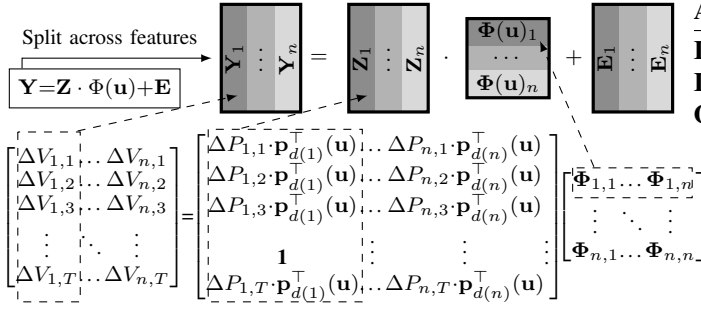


Fig. 1. Multivariate linear regression model

B. Data transformation with multiplicative randomization

To calculate $\mathbf{Z}_i^T \mathbf{W} \mathbf{Z}_j$ and $\mathbf{Z}_i^T \mathbf{W} \mathbf{Y}_j$, $i \neq j$, without confidential (or personal) data disclosure, we adapted the protocol from [21]. This approach randomizes (encrypts) data through linear algebra, in such a way that both data and coefficients are protected when performing a linear regression. In what follows a description of these transformations is provided.

Multiplicative randomization of the data consists of multiplying the data matrix $\mathbf{X}$ by invertible perturbation matrices. If the perturbation matrix $\mathbf{M}$ pre-multiplies $\mathbf{X}$, i.e., $\mathbf{MX}$, the records are randomized. On the other hand, if the perturbation matrix $\mathbf{Q}$ post-multiplies $\mathbf{X}$, i.e., $\mathbf{XQ}$, then the features are randomized, and consequently the coefficients.

1) *Features randomization*: since the problem formulated in Section II has $\mathbf{X}$ divided by features, we can construct $\mathbf{Q}$ as a diagonal matrix, where diagonal matrices, $\mathbf{Q}_i$, are privately defined by agent i . This structure allows each agent to locally compute $\mathbf{X}_i \mathbf{Q}_i$, without sharing $\mathbf{Q}_i$,

$$\underbrace{[\mathbf{X}_1, \dots, \mathbf{X}_n]}_{=\mathbf{X}} \underbrace{\begin{bmatrix} \mathbf{Q}_1 & \mathbf{0} \\ & \ddots \\ \mathbf{0} & \mathbf{Q}_n \end{bmatrix}}_{=\mathbf{Q}} = [\mathbf{X}_1 \mathbf{Q}_1, \dots, \mathbf{X}_n \mathbf{Q}_n]. \quad (10)$$

2) *Records randomization*: since all elements of j^{th} column of $\mathbf{M}$ multiplies all elements of j^{th} row in $\mathbf{X}$, it is not possible to apply the previous reasoning when defining $\mathbf{M}$. This limitation motivated the definition of $\mathbf{M}$ as a product of n private matrices $\mathbf{M}_i$, $i \in \{1, \dots, n\}$.

The biggest advantage is to avoid relying on a third party to define $\mathbf{M}$. This protocol ensures that no agents know $\mathbf{M}$, but all contribute to define it, as described in Algorithm 1. The privacy of this protocol depends on r , which is chosen according to the number of unique values on $\mathbf{X}_i$ [21].

C. Collaborative calculation of sensitivity factors

With the objective of fitting the model from (3) without data disclosure, the protocol described in Algorithm 1 is applied to transform $\mathbf{Z}$ and $\mathbf{Y}$ in such a way that: (i) the estimated coefficients are a secret transformation of the original ones, (ii) agents are unable to recover the private data through the

Algorithm 1 Data Encryption.

Input from i^{th} agent: $\mathbf{X}_i \in \mathbb{R}^{T \times s}$ and $\mathbf{M}_i \in \mathbb{R}^{T \times T}$

Input from j^{th} agent ($j \neq i$): $\mathbf{M}_j \in \mathbb{R}^{T \times T}$

Output: $\mathbf{MX}_i = \mathbf{M}_1 \dots \mathbf{M}_n \mathbf{X}_i$

- 1: **Initialization:** Agent i generates random $\mathbf{C}_i \in \mathbb{R}^{T \times (r-s)}$, invertible $\mathbf{D}_i \in \mathbb{R}^{r \times r}$, and shares $\mathbf{F}_i$ with the n^{th} agent,

$$\mathbf{F}_i = [\mathbf{X}_i, \mathbf{C}_i] \mathbf{D}_i. \quad (11)$$

- 2: Agent n gets $\mathbf{F}_i$ and shares $\mathbf{M}_n \mathbf{F}_i$ with $(n-1)^{th}$ agent.
- 3: **for** agent $j = n-1, \dots, 1$ **do**
- 4: Agent j receives $\left(\prod_{k=j+1}^n \mathbf{M}_k \right) \mathbf{F}_i$, and
- 5: **if** $j > 1$: shares $\mathbf{M}_j \left(\prod_{k=j+1}^n \mathbf{M}_k \right) \mathbf{F}_i$ with agent $j-1$
- 6: **else:** shares $\mathbf{M}_j \left(\prod_{k=j+1}^n \mathbf{M}_k \right) \mathbf{F}_i$ with agent i
- 7: **end for**
- 8: Agent i gets $\mathbf{MF}_i$ from the 1^{st} agent and recovers $\mathbf{MX}_i$,

$$[\mathbf{MX}_i, \mathbf{MC}_i] = \mathbf{MF}_i \mathbf{D}_i^{-1}. \quad (12)$$

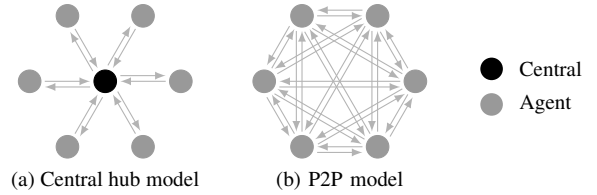


Fig. 2. Two communication schemes

exchanged information, and (iii) cross-correlations cannot be obtained, i.e., agents are unable to recover $\mathbf{Z}^T \mathbf{Z}$ nor $\mathbf{Z}^T \mathbf{Y}$.

The optimization problem with transformed data,

$$\hat{\Phi} = \underset{\Phi}{\operatorname{argmin}} \frac{1}{2} \|\mathbf{MW}^{\frac{1}{2}} \mathbf{Y} - \mathbf{MW}^{\frac{1}{2}} \mathbf{ZQ}\Phi\|_2^2, \quad (13)$$

has the multivariate least-squares solution

$$\hat{\Phi} = (\mathbf{Q}^T \mathbf{Z}^T \mathbf{W}^{\frac{1}{2}} \mathbf{M}^T \mathbf{MW}^{\frac{1}{2}} \mathbf{ZQ})^{-1} (\mathbf{Q}^T \mathbf{Z}^T \mathbf{W}^{\frac{1}{2}} \mathbf{M}^T \mathbf{MW}^{\frac{1}{2}} \mathbf{Y}), \quad (14)$$

which preserves the relation between the original data if $\mathbf{M}$ is orthogonal, i.e., $\mathbf{M}^T \mathbf{M} = \mathbf{I}$. However, such property makes possible to compute cross-correlations.

This limitation is handled by using $\mathbf{Q}^T \mathbf{Z}^T \mathbf{W}^{\frac{1}{2}} \mathbf{M}^{-1}$ instead of $\mathbf{Q}^T \mathbf{Z}^T \mathbf{W}^{\frac{1}{2}} \mathbf{M}^T$ in (14). We propose the use of Algorithm 1 to compute $\mathbf{MW}^{\frac{1}{2}} \mathbf{Z}_i \mathbf{Q}_i$, $\mathbf{Q}_i^T \mathbf{Z}_i^T \mathbf{W}^{\frac{1}{2}} \mathbf{M}^{-1}$ and $\mathbf{MW}^{\frac{1}{2}} \mathbf{Y}_i$.

The coefficients are then computed by a central agent, or in a P2P scheme, as illustrated in Fig. 2. It is important to emphasize that the solution (i.e., smooth functions of the coefficients) obtained with the central and P2P scheme is exactly the same and only the flow of information exchange between the different data owners is modified.

IV. CASE-STUDY

A. Description

The proposed method was tested in a 33-node typical Portuguese LV grid (Fig. 3) [22] and in two UK grids from the

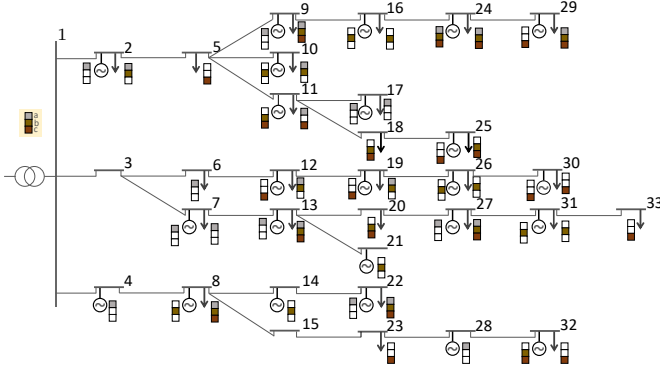


Fig. 3. 33-node test system

“Low Voltage Network Solutions” project [23], hereby referred as *grid_33N*, *grid_N1F2* and *grid_N1F3*, respectively.

The 30-min load time series of the first test system, *grid_33N*, comes from an Irish smart metering trial [24], which were then combined with PV generation measurements from [25], and with EV charging data from [26], comprising 53 single-phase customers. An unbalanced power flow algorithm, based on the backward-forward sweep method, was applied with this data to generate the corresponding 22000 records of “measured” voltage magnitude in each LV node.

For the other two grids, the consumption and generation profiles were gathered from the “Customer-Led Network Revolution” project [27], for which 5720 power flows were run to estimate the voltage profiles. *grid_N1F2* has 31 single-phase loads spread over 111 three-phase nodes, while *grid_N1F3* has 39 single-phase loads and 123 three-phase nodes.

Since the power flow analysis requires a reference voltage, the voltages of each SS were obtained by running previously a power flow analysis with a generic MV grid [28], in which the LV loads from the test systems were allocated to secondary substations.

In order to simulate the collaborative analytics framework described in Section III, each LV consumer is referred as an agent (data owner) and the DSO assumes the role of central agent. In this configuration, one may assume that consumers are not willing to share their private consumption/generation data but trust the DSO to compute the final sensitivity factors. Therefore, the DSO: a) receives from each agent the encrypted matrices $\mathbf{MW}^{\frac{1}{2}} \mathbf{Z}_i \mathbf{Q}_i$, $\mathbf{Q}_i^T \mathbf{Z}_i^T \mathbf{W}^{\frac{1}{2}} \mathbf{M}^{-1}$, $\mathbf{MW}^{\frac{1}{2}} \mathbf{Y}_i$ and $\mathbf{Q}_i$, which are obtained by Algorithm 1; b) computes the encrypted sensitivity factors according to (14); and c) decrypts, using $\mathbf{Q}_i$, to obtain the original sensitivity factors.

As mentioned before, the results with the P2P scheme would be the same, but without the DSO as the central agent.

B. Quality of the sensitivity factors estimation

The historical data for *grid_33N* was divided in 320 days for model fitting (sensitivity factors learning) and 120 days for testing; for the two UK grids the division was 80 and 35 days respectively. The voltage and active power variations needed

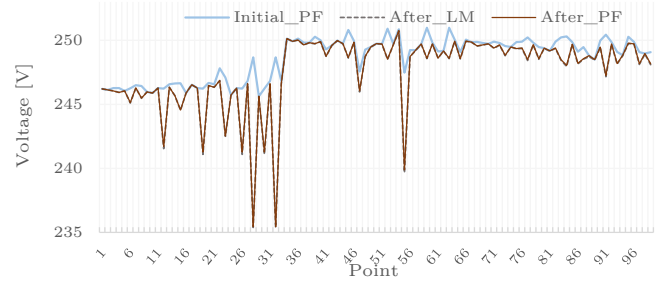


Fig. 4. Voltage after disturbing the active power injection/consumption by +14 kW, +11 kW, +7 kW, +6 kW, +10 kW and +7 kW, simultaneously, in points 19, 28, 52, 55, 79, and 91, respectively.

for the regression model were obtained from the differences between two consecutive records. Finally, the quality of the regression was assessed by comparing the results of the linear model against the power flow results, using the mean absolute error (MAE), the root-mean-squared error (RMSE) and the mean absolute percentage error (MAPE) (Table I).

TABLE I
QUALITY OF THE REGRESSION FOR THE DIFFERENT TEST GRIDS, USING CONSTANT SENSITIVITY FACTORS

	<i>grid_33N</i>	<i>grid_N1F2</i>	<i>grid_N1F3</i>
MAE [V]	0.00955	0.00289	0.00520
RMSE [V]	0.01621	0.00463	0.00906
MAPE [%]	0.00383	0.00118	0.00215

In overall, the constant sensitivity factors allow to determine the new voltages with high accuracy. The worse results for *grid_33N* may be related to the higher consumption values and the wider range of voltages in the SS. For instance, while voltage in the SS of *grid_33N* ranges roughly between 220V and 262V, in the UK grids it varies between 240V and 251V.

In order to validate even further the generalization of the model, 1000 records of historical data were randomly chosen to apply multiple random active power control set-points between 5 and 15 kW across the *grid_33N*. The new voltages estimated by the linear sensitivities were then compared to the results obtained with the power flow calculation. The MAE, RMSE and MAPE are now 0.03347V, 0.09243V and 0.01361%, respectively, which are higher than in Table I most likely due to the larger active power changes. Fig. 4 illustrates, for one of these 30-min periods, the voltages before and after the disturbances (“Initial_PF”), showing the similarity of the voltage states given by the linear model (“After_LM”) to the ones obtained with the power flow (“After_PF”).

C. Results with conditional sensitivity factors

In a first approach to build varying coefficients, the active power flow in the SS, from the same phase as the customer (easily obtained from the regression model as described in Section II), was selected as the external variable to condition the sensitivity factors. Then, since the method requires to define how local the model is, a convenient value of α has to be found (*Test#1*). However, the degree of the polynomial that represents the external variable around each fitting point

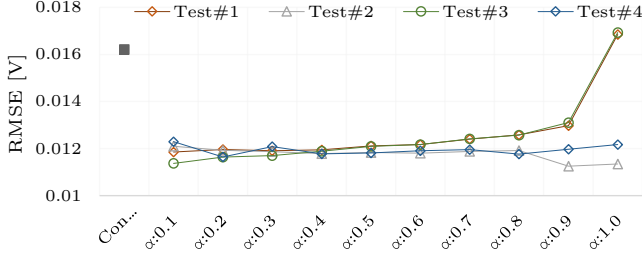


Fig. 5. RMSE comparison between the constant and the conditional model.

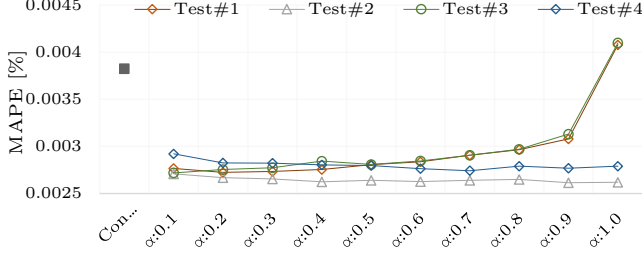


Fig. 6. MAPE comparison between the constant and the conditional model.

has also to be selected, under the trade-off between precision and computation time (*Test#2*). The same trade-off arises when choosing the number of fitting points for the external variable (*Test#3*).

Lastly, besides finding a suitable configuration for the model, it is also of interest to test other external variables. Considering the possibility of solving voltage violations in real-time operation, and assuming that the SS would have its voltage monitored, the active power flow at the SS was replaced by the voltage, from the same phase as the customer, for external variable (*Test#4*).

Fig. 5 and 6 compare the use of constant and variable sensitivity factors for *Test#1-4*, considering the RMSE and MAPE. Despite the already good accuracy of the constant model, there is a clear performance improvement when using the conditional model.

In terms of computational time, *Test#1* took on total 452 minutes (for each α value, roughly 2.3-min per fitting point) to run a C++ implementation in a 64 bits Intel® Xeon® E3-1240 v6, 32GB RAM. The same ratio holds in other tests when using a 0-degree polynomial, and doubles with a 1-degree polynomial (i.e., 4.5-min per fitting point).

Looking at the different α values, a 0-degree polynomial tends to perform better for low values (more local model), while a first degree polynomial reveals to be less sensitive to the chosen value. Comparing *Test#1* and *Test#3* it is possible to conclude that an higher number of *fitting points* does not lead to better results, with the disadvantage of doubling the execution time. Finally, for this dataset, the use of the secondary substation voltage as an external variable revealed to be a slightly better option.

Finally, Fig. 7 illustrates, for *Test#1* and two grid points, how the sensitivity factor changes as the grid loading increases (i.e., MV/LV substation load as a proxy).

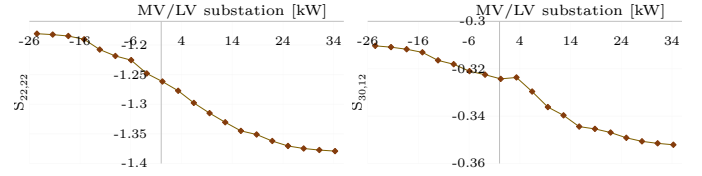


Fig. 7. Two voltage sensitivities from *grid_33N* with respect to the SS active power flow.

D. Solving voltage violations

This subsection presents one potential application of the sensitivity factors, which integrates them in a real-time LV control function. For instance, every 30 minutes a state estimator similar to [1] would reconstruct the most likely state of the system (only the estimation of missing voltage measurements would be required), supported by a minimum number of smart meters with real-time communication. In face of a violation, whether of ground-to-neutral limits or in phase unbalancing limits [29], the objective would be to define control set-points for active power flexibility, in each point, so these violations could be solved in the most economic way. A possible formulation for this optimization problem would be:

$$\min J = \sum_{i=1}^n \Delta P_i^{up} \cdot C_i^{up} - \Delta P_i^{down} \cdot C_i^{down} \quad (15)$$

subject to

$$\begin{cases} 0 \leq \Delta P_i^{up} \leq \Delta P_i^{max} \\ \Delta P_i^{min} \leq \Delta P_i^{down} \leq 0 \end{cases} \quad (16)$$

$$V_i^{min} - V_i^{old} \leq \Delta V_i \leq V_i^{max} - V_i^{old} \quad (17)$$

$$\frac{\Delta V_k^{max}}{V_k^{avg}} \times 100 \leq l \quad (18)$$

$$V_k^{avg} = \frac{V_{a,k} + V_{b,k} + V_{c,k}}{3} \quad (19)$$

$$\Delta V_P^{max} = \max\{|V_{a,k} - V_k^{avg}|, |V_{b,k} - V_k^{avg}|, |V_{c,k} - V_k^{avg}|\} \quad (20)$$

$$\Delta V_i = \sum_{j=1}^n (\Delta P_j^{up} - \Delta P_j^{down}) \cdot S_{i,j} \quad (21)$$

where ΔP_i^{up} and ΔP_i^{down} are the amount of active power flexibility (decrease or increase operating point), respectively, at the connection point i , and C_i^{up} and C_i^{down} are the corresponding costs. Any power variation is bounded by ΔP_i^{min} and ΔP_i^{max} , while the ground-to-neutral voltages admit a minimum value at V_i^{min} and a maximum at V_i^{max} . Additionally, the highest voltage unbalance (IEEE definition in [29]) for node k , among its ground-to-neutral voltages $V_{a,k}$, $V_{b,k}$, and $V_{c,k}$, must be less than the value set by l (e.g., 2%).

To test and demonstrate this formulation, *grid_33N* grid was considered again and assumed that a flexibility of ± 2.5 kW from each customer was available. Then, considering one of the 30-min historical snapshots, two tests were performed: 1) set a maximum value of 250V for all the ground-to-neutral

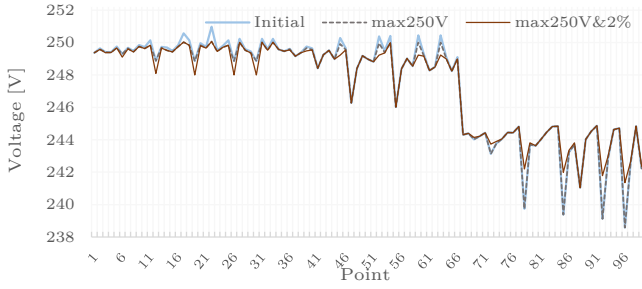


Fig. 8. Voltage per connection point for: (i) initial voltages; (ii) voltage after imposing $\leq 250V$ limit; (iii) voltage after 2% phase unbalance limit.

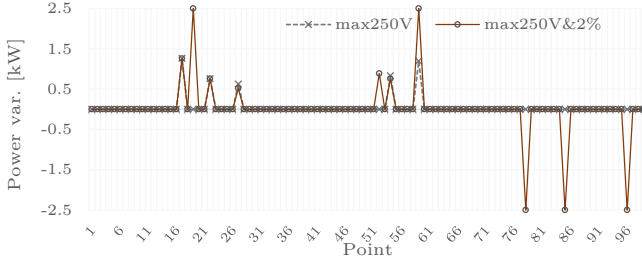


Fig. 9. Power variation after: (i) $\leq 250V$ limit; (ii) 2% phase unbalance limit.

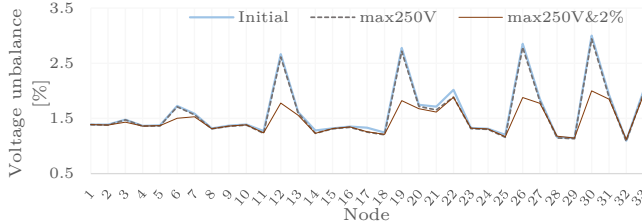


Fig. 10. Unbalance per node: (i) initial; (ii) after imposing $\leq 250V$ limit; (iii) after 2% phase unbalance limit.

voltages, and 2) also force a maximum value of 2% for the voltage unbalance. Fig. 8 depicts the initial voltages of all the connection points, the voltages considering the 250V constraint and after adding the 2% constraint. In this example, the over-voltages (above 250V) were solved by increasing the injected power at certain points, as shown in Fig. 9. However, looking at Fig. 10 one can see that the initial unbalance was high and this first correction did little to improve it. Once the 2% limit was enforced, the optimization algorithm had to use the sensitivity factors not only to reduce voltages above 250V, but also to seek a control solution that would ensure a better phase balancing. This resulted in several additional net-load changes, some to decrease and others to increase as depicted in Fig. 9 for curve *max250V&2%*. To finalize, it should be noted that the voltage profiles displayed in Fig. 8 are the result of a power flow calculation, proving the adequacy of the sensitivity factors as a proxy of the complete grid model.

E. Computing operating envelopes

With the growing presence of controllable devices at the edge of the distribution grid (e.g., EVs, PVs and battery storage units), a number of use cases have been proposed for synergies that benefit both the consumer and the grid. However, the LV grid management has been oversimplified

(mainly due to limited monitoring and control capabilities) by the use of limits for power injection imposed by regulation, discarding the time-varying state of the grid, e.g., in some periods it would be possible for a prosumer to inject more power, while in others it should be restricted to avoid technical problems. The concept of operating envelopes tackles the drawbacks of fixed limits by defining time-varying limits at the individual LV consumer level, and that can be aggregated up to the secondary substation level [30], which can be computed by modifying the objective function of the formulation in IV-D.

In order to demonstrate the concept of operating envelopes, let us consider one of the operating snapshots for the *grid_33N* grid and, after defining limits for voltage values and maximum export/import active powers, compute for each consumer its operating envelope. Since this LV system has very few consumers please bear in mind that the boundaries were chosen to facilitate the demonstration: 255V/235V as maximum/minimum voltage and +15/-15 kW as maximum/minimum net-load. Fig. 11 shows the maximum and the minimum net-load allowed to each consumer connected to phase c, after running the optimization for each one of them individually (i.e., keeping the remaining net loads fixed). As expected, consumers connected near the head of the feeder have little influence over the voltages, which are mainly imposed by the MV grid, and hence the large envelopes. On the other hand, consumers located near the end of the feeder are more likely to see their flexibility range constrained.

Another interesting exercise would be to find the operating envelope for the secondary substation (i.e., min and max range of the transformer net-load) by computing the required flexibility from each consumer that comes from the maximization of the total flexibility exchanged between the MV and the LV grid. To demonstrate this, considering the same period as above and the same constraints, the objective function of the optimization problem was changed to maximize first the active power flow from the MV grid to the LV grid, and then to maximize the opposite flow. The corresponding net-load setpoints from consumers in phase c are represented in Fig. 12, totaling a maximum of 169.8 kW from the MV to the LV grid and 190.7 kW in the opposite direction, which is higher since the initial voltages are closer to the lower limit (see Fig. 13). Also, it is visible that the flexibility provided by consumers at the end of the feeder can be greatly constrained. To balance these flexibility ranges per consumer, the objective function could be enhanced with an additional term like the variance of the ranges. Finally, the activation of such flexibility would lead to the voltages depicted in Fig. 13, with curves *min* and *max* corresponding to the maximization of the power exchange from the MV to the LV and vice-versa, respectively.

V. CONCLUSIONS

This work showed that accurate sensitivity factors can be extracted from smart meter data, while keeping this data private, and embedded in predictive and real-time voltage control functions for LV grids where topology and electrical parameters are not fully characterized or have erroneous

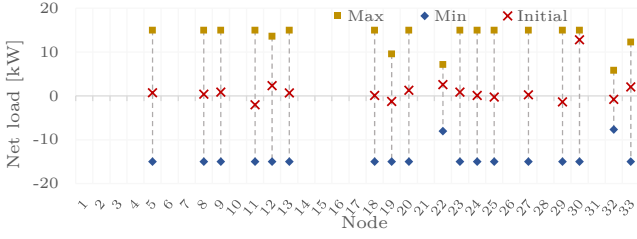


Fig. 11. Independent envelopes for consumers connected to phase c.

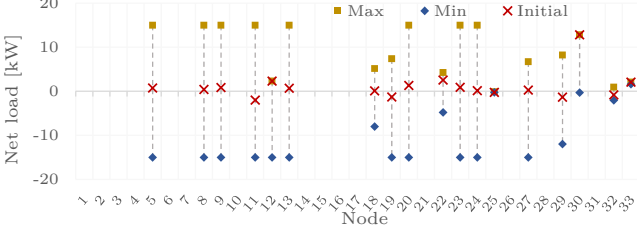


Fig. 12. Net-load setpoints for the consumers connected to phase c that maximize the MV/LV power exchange (operating envelope).

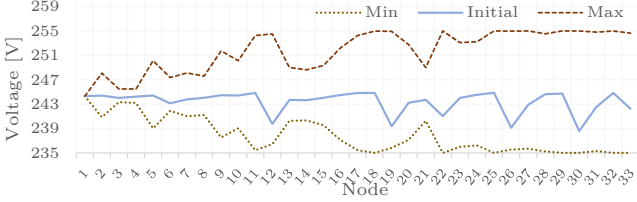


Fig. 13. Initial voltages per node of phase c and the resulting voltages considering the limits of the MV/LV substation envelope.

information. Moreover, the use of varying coefficients in the linear regression framework accurately captured the changes in sensitivity factors due to varying grid operating conditions.

Nonetheless, some limitations in the proposed data-driven method can be pointed out for practical applications. Firstly, smart meter measurements (active power and voltage magnitude) from all LV nodes are needed. Reducing the number of required measurements can be further investigated, which is expected to further decrease the communication requirements and improve replicability. Secondly, the data quality will also affect performance. Nevertheless, in both cases, LV state estimation algorithms (see [1] for instance) can mitigate these impacts.

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